

Coherent Logic — an overview

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Crash course in Coherent Logic (CL)

- Basics

- Proof theory for CL

- Metatheory

- Translation from FOL to CL

- Evaluation of CL as a fragment of FOL

Automated reasoning in CL

- Automated reasoning

- Elimination of function symbols

- Proof search strategies

Applications of CL

- Proof assistants

- Model finding

- Constructive algebra

Coherent logic preliminaries 1

- ▶ Fix a finite first-order signature Σ
- ▶ Positive formulas: built up from atoms using $\top, \perp, \vee, \wedge, \exists$
- ▶ Coherent implications (sentences): $\forall \vec{x}. (C \rightarrow D)$ with C, D positive formulas
- ▶ Coherent theory: axiomatized by coherent sentences
- ▶ $\vee\exists\wedge$ -formula: $(\exists \vec{y}_1.A_1) \vee \cdots \vee (\exists \vec{y}_k.A_k)$, $k \geq 0$, with each A_i a (possibly empty) conjunction of atoms
- ▶ Lemma 1. Every positive formula is (constructively) equivalent to a $\vee\exists\wedge$ -formula. Proof by induction:
 - ▶ Base cases: atom (one disjunct, empty $\exists$, one conjunct); $\perp$ (empty $\vee$); $\top$ (one disjunct with empty $\exists, \wedge$)
 - ▶ Induction cases: $\vee$ (trivial); $\wedge$ (distributivity + $(\exists x.\varphi) \wedge (\exists y.\psi)$ iff $\exists xy. (\varphi \wedge \psi)$); $\exists$ (commutes with $\vee$)

Coherent logic preliminaries 2

- ▶ Lemma 2. Every coherent implication is (constructively) equivalent to a finite set of coherent implications $\forall \vec{x}. (C \rightarrow D)$ with C a conjunction of atoms and D a $\forall\exists\wedge$ -formula
- ▶ Proof. Use Lemma 1 to replace C and D by $\forall\exists\wedge$ -formulas. Then use $(\varphi \vee \psi) \rightarrow D$ iff $(\varphi \rightarrow D) \wedge (\psi \rightarrow D)$, and $(\exists y. \varphi) \rightarrow D$ iff $\forall y. (\varphi \rightarrow D)$
- ▶ Notation: we use the format of Lemma 2, leaving out the universal prefix, and omitting the premiss ' $C \rightarrow$ ' if $C \equiv \top$
- ▶ Discuss: $\exists y. \top$ and $\exists y. \perp$ and $\forall y. \top$ and $\forall y. \perp$
- ▶ Full compliance with Tarski semantics if Σ has a constant

Examples

- ▶ all usual equality axioms, including congruence
- ▶ $p \vee np$ and $p \wedge np \rightarrow \perp$ (NB $p \vee \neg p$ is **not** coherent)
- ▶ lattice theory: $\exists z. \text{meet}(x, y, z)$
- ▶ geometry: $p(x) \wedge p(y) \rightarrow \exists z. \ell(z) \wedge i(x, z) \wedge i(y, z)$
- ▶ rewriting, $\diamond$ -property: $r(x, y) \wedge r(x, z) \rightarrow \exists u. r(y, u) \wedge r(z, u)$
- ▶ weak-tc-elim: $r^*(x, y) \rightarrow (x = y) \vee \exists z. r(x, z) \wedge r^*(z, y)$
- ▶ seriality: $\exists y. s(x, y)$ (who needs a function?)
- ▶ field theory: $(x = 0) \vee \exists y. (x \cdot y = 1)$
- ▶ local ring: $\exists y. (x \cdot y = 1) \vee (\exists y. ((1 - x) \cdot y = 1))$ (equivalent to the more common: if $x + y$ is a unit, then x is a unit or y is a unit).

History of CL

- ▶ Skolem (1920s): coherent formulations of lattice theory and projective geometry, calling the axioms "Erzeugungsprinzipien" (production rules), anticipating ground forward reasoning. Using CL,
 - ▶ Skolem solved a decision problem in lattice theory
 - ▶ Skolem gave a method to test in/dependence from the axioms of plane projective geometry (example: Desargues' Axiom)
- ▶ Grothendieck (1960s): geometric morphisms preserve geometric logic (= coherent logic + infinitary disjunction). This is quite complicated, but we'll see a glimpse in the forcing semantics of coherent logic given later.

A proof theory for CL

- ▶ In short: ground forward reasoning with case distinction and introduction of witnesses (ground tableau reasoning)
- ▶ In full: define inductively $\Gamma \vdash_{\vec{y}}^T A$, where A (Γ) atom (set of atoms) with all variables in $\vec{y}$, in case
 - (base) A is in Γ , or in case
 - (step) T has an axiom $\forall \vec{x}. (C \rightarrow (\exists \vec{y}_1.B_1) \vee \dots \vee (\exists \vec{y}_k.B_k))$ such that for some sequence of terms $\vec{t}$ with variables in $\vec{y}$ we have
 - ▶ $C[\vec{t}/\vec{x}]$ is a subset of Γ , and
 - ▶ $\Gamma, B_i[\vec{t}/\vec{x}] \vdash_{\vec{y}, \vec{y}_i}^T A$ for all $i = 1, \dots, n$ (NB $\vec{y}_i$ fresh wrt $\vec{y}$)
- ▶ Rough visualization as a tree with inner nodes like

$$\frac{\Gamma, B_1[\vec{t}/\vec{x}] \quad \dots \quad \Gamma, B_n[\vec{t}/\vec{x}]}{\Gamma} \text{ axiom}$$

- ▶ NB we omit conclusion A in all the nodes, but we should actually keep track of the $\vec{y}, \vec{y}_i$. Looking ahead, pairs like $(\vec{y}; \Gamma)$ will be the forcing conditions, $\approx$ finite Kripke worlds.

Derivation trees in CL, example and general procedure

- ▶ Let T consists of $p \vee \exists x. q(x)$ and $p \rightarrow \perp$ and $q(y) \rightarrow r$
- ▶ Derivation tree for $\emptyset \vdash_{\emptyset}^T r$

$$\frac{\frac{(\perp)}{\{p\}} p \rightarrow \perp \quad \frac{\{q(c), r\}}{\{q(c)\}} q(y) \rightarrow r}{\emptyset} p \vee \exists x. q(x)$$

- ▶ Tree construction: from $\emptyset$, repeat exhaustively 1,2,3 below
 1. Pick a leaf node ($\neq (\perp)$) without A in its Γ (else done)
 2. Pick **fairly a Γ -false instance** of an axiom of T (else fail: Γ is a model of T not containing A , so A is undervivable)
 3. Extend the tree in the leaf node according to the instance
- ▶ Fairness is tricky to define, but crucial for the following completeness result (to be proved on the next slide):
- ▶ The tree construction stops in 1 iff A is derivable (**if-part!**)
- ▶ Example for explaining un/fairness: $\exists y.s(x, y)$ and $p(0)$

Soundness and completeness wrt Tarski semantics

- ▶ Soundness easily proved by induction on $\Gamma \vdash_{\vec{y}}^T A$
- ▶ **Not** complete: $\emptyset \vdash_{\emptyset}^{\forall x. \perp} p$ underivable without a constant in Σ
- ▶ Silly, let's assume a constant in Σ , or just $\exists x. \top$
- ▶ Proof of completeness: essentially non-constructive.
Assume $\forall \vec{y}. (\Gamma \rightarrow A)$ holds in any model of T . Build the tree for $\Gamma \vdash_{\vec{y}}^T A$. Recall that the sets Γ grow along the branches. If the tree is finite, it is a proof (2 cannot happen). If not, it has an infinite branch by König's Lemma. Collect the set of variables Y and the set of atoms M along the infinite branch. Build a model with domain $\text{Tm}^{\Sigma}(Y)$ and positive diagram M . This is a model of T (**by fairness**) containing Γ but not A . Contradiction.
- ▶ Proof theory easily extended to arbitrary coherent conclusions of a coherent theory T .

Metatheoretic results and remarks

- ▶ Corollary of completeness: given a coherent theory T , classically provable coherent sentences are constructively provable
- ▶ For **geometric** logic this is called Barr's Theorem (anticipated by Lawvere and Deligne)
- ▶ Completeness and Barr's Theorem are **not** constructive
- ▶ Barr's Theorem for **coherent** logic can be proved constructively using a cut-elimination argument
- ▶ Coherent completeness wrt forcing semantics is constructively provable, but does not give the conservativity of classical reasoning
- ▶ NB: the forcing semantics is sound wrt constructive logic for arbitrary formulas

Translation from FOL to CL

- ▶ Idea: introduce two new predicate symbols $T(\psi)$, $F(\psi)$ for each subformula ψ of a given formula φ , with the arities of $T(\psi)$, $F(\psi)$ being the number of free variables of ψ . The rules for signed tableaux are coherent axioms:
 - ▶ if $\psi(\vec{x}) \equiv \psi_1 \wedge \psi_2$, then $\begin{cases} T(\psi)(\vec{x}) \rightarrow T(\psi_1)(\vec{x}) \wedge T(\psi_2)(\vec{x}) \\ F(\psi)(\vec{x}) \rightarrow F(\psi_1)(\vec{x}) \vee F(\psi_2)(\vec{x}) \end{cases}$
 - ▶ if $\psi(\vec{x}) \equiv \psi_1 \vee \psi_2$, then ...
 - ▶ if $\psi(\vec{x}) \equiv \psi_1 \rightarrow \psi_2$, then $\begin{cases} T(\psi)(\vec{x}) \rightarrow F(\psi_1)(\vec{x}) \vee T(\psi_2)(\vec{x}) \\ F(\psi)(\vec{x}) \rightarrow T(\psi_1)(\vec{x}) \wedge F(\psi_2)(\vec{x}) \end{cases}$
 - ▶ if $\psi(\vec{x}) \equiv \neg\psi_1$, then ...
 - ▶ if $\psi(\vec{x}) \equiv \forall y. \psi_1(\vec{x}, y)$, then $\begin{cases} T(\psi)(\vec{x}) \rightarrow T(\psi_1)(\vec{x}, y) \\ F(\psi)(\vec{x}) \rightarrow \exists y. F(\psi_1)(\vec{x}, y) \end{cases}$
 - ▶ if $\psi(\vec{x}) \equiv \exists y. \psi_1(\vec{x}, y)$, then $\begin{cases} T(\psi)(\vec{x}) \rightarrow \exists y. T(\psi_1)(\vec{x}, y) \\ F(\psi)(\vec{x}) \rightarrow F(\psi_1)(\vec{x}, y) \end{cases}$
 - ▶ if $\psi(\vec{x})$ is atomic, then $(T(\psi)(\vec{x}) \wedge F(\psi)(\vec{x})) \rightarrow \perp$
- ▶ By the completeness of signed tableaux: φ is a tautology iff $F(\varphi) \vdash_{\emptyset}^{Coh(\varphi)} \perp$, with $Coh(\varphi)$ all the above axioms

Example in propositional logic: Peirce's Law

- ▶ Peirce's Law: $\varphi \equiv ((p \rightarrow q) \rightarrow p) \rightarrow p$
- ▶ To prove: $F(((p \rightarrow q) \rightarrow p) \rightarrow p) \vdash_{\emptyset}^{Coh(\varphi)} \perp$
- ▶ Part of $Coh(\varphi)$ that is actually used:
 1. $F(((p \rightarrow q) \rightarrow p) \rightarrow p) \rightarrow (T((p \rightarrow q) \rightarrow p) \wedge F(p))$
 2. $T((p \rightarrow q) \rightarrow p) \rightarrow (F(p \rightarrow q) \vee T(p))$
 3. $F(p \rightarrow q) \rightarrow (T(p) \wedge F(q))$
 4. $(T(p) \wedge F(p)) \rightarrow \perp$
- ▶ Proof: use 1, 2, 3 and split on $F(p \rightarrow q) \vee T(p)$, ...
- ▶ Details on the blackboard
- ▶ Proof of φ : take $T \equiv \lambda\varphi. \varphi$, $F \equiv \lambda\varphi. \neg\varphi$. Then 1,2,3,4 are easy (but classical), the CL proof is also a proof in propositional logic, and we finish by RAA

Example in predicate logic: the Drinker's Paradox

- ▶ Drinker's Paradox: $\varphi \equiv \exists x. (d(x) \rightarrow \forall y.d(y))$
- ▶ To prove: $F(\exists x. d(x) \rightarrow \forall y.d(y)) \vdash_{\emptyset}^{Coh(\varphi)} \perp$
- ▶ Part of $Coh(\varphi)$ that is actually used:
 1. **no** take-off without $\exists x.\top$, alternative: prove $F(\varphi) \vdash_{\{c\}}^{Coh(\varphi)} \perp$
 2. $\forall x. (F(\exists x. d(x) \rightarrow \forall y.d(y))) \rightarrow F(d(x) \rightarrow \forall y.d(y))$
 3. $\forall x. (F(d(x) \rightarrow \forall y.d(y)) \rightarrow (T(d(x)) \wedge F(\forall y.d(y))))$
 4. $F(\forall y.d(y)) \rightarrow \exists y.F(d(y))$
 5. $\forall x. (T(d(x)) \wedge F(d(x))) \rightarrow \perp$
- ▶ Proof: use 1 and get c , instantiate 2 and 3 with c and get $T(d(c)) \wedge F(\forall y.d(y))$, so by 4 we get c' with $F(d(c'))$, ...
- ▶ Details on the blackboard
- ▶ Proof of φ in FOL: take $T \equiv \lambda\varphi. \varphi$, $F \equiv \lambda\varphi. \neg\varphi$. Then 1–5 are easy (Tarski and classical), the CL proof is also a proof in FOL, and we finish by RAA

Translation from FOL to CL (ctnd)

- ▶ Skolem (1920): Every FOL theory has a definitional extension that is equivalent to a $\forall\exists$ theory
- ▶ Many variations possible (Polonsky, Dyckhoff & Negri, Fisher, Mints)
- ▶ Possible objectives: fewer new predicates, fewer CL axioms ..., keeping a coherent axiom coherent
- ▶ Polonsky proposed several improvements, starting from NNF, flipping polarities, also using reversed tableaux rules
- ▶ Dyckhoff & Negri: add $T(\psi)(\vec{x}) \rightarrow \psi(\vec{x})$ and $(F(\psi)(\vec{x}) \wedge \psi(\vec{x})) \rightarrow \perp$ for all atomic ψ and obtain: Every FOL theory has a positive semi-definitional extension that is equivalent to a CL theory
- ▶ Consequences in CL are always constructive
- ▶ Translation of FOL to CL contains many non-constructive steps, often more than necessary

Evaluation of CL as a fragment of FOL

- ▶ Constructive, with classical logic a conservative extension
- ▶ Simpler metatheory: proof theory, completeness, conservativity of skolemization (elimination of $\exists$)
- ▶ Applications to metamathematics: independence, decision problems
- ▶ Other applications:
 - ▶ automated reasoning, supporting proof assistants
 - ▶ model finding
 - ▶ constructive algebra

Automated reasoning (AR)

- ▶ We focus on AR in (fragments of) FOL
- ▶ There are dozens of FOL provers (Vampire wins CASC)
- ▶ TPTP is a large database of AR problems (CNF/FOL/HOL)
- ▶ Different branch of AR: model finding (SAT/CNF/FMT)
- ▶ There are a handful of CL provers (competitive on CL problems, but not on FOL problems):
 - ▶ SATCHMO+ (Bry et al.)
 - ▶ Argo, Larus (Janicic et al.)
 - ▶ Geo (Nivelle et al.)
 - ▶ Colog (Fisher)
 - ▶ EYE (De Roo, semantic web)
- ▶ Most CL provers support only 0-ary function symbols
- ▶ We describe later how to eliminate function symbols

Rationale for automated reasoning in CL

- ▶ Expressivity of CL is between CNF (resolution) and FOL
- ▶ Different balance: expressivity versus efficiency
- ▶ Skolemization (elimination of $\exists$) not necessary
 - ▶ Skolemization makes the Herbrand Universe infinite
 - ▶ Why skolemize an axiom like $p(x, y) \rightarrow \exists z. p(x, z)$?
 - ▶ Skolemization changes meaning (problematic for interactive theorem proving, and for obtaining proof objects)
 - ▶ Skolem functions obfuscate symmetries (cf. $\diamond$ -property)
 - ▶ **But:** skolemized proofs can be exponentially shorter!
- ▶ Ground forward reasoning is very simple and intuitive, proof objects can easily be obtained
- ▶ **But:** non-ground proofs can be exponentially shorter!

Elimination of function symbols

- ▶ Idea: use the graph instead of the function, i.e., new $(n+1)$ -predicates for n -ary functions, for example:
 - ▶ For constants: $c(x)$ (expressing $c = x$), axiom $\exists x. c(x)$
 - ▶ For unary functions: $s(x, y)$ (expressing $s(x) = y$), axiom $\exists y. s(x, y)$
- ▶ Example: the term $f(s(x), o)$ leads to a condition $s(x, y) \wedge o(u) \wedge f(y, u, z)$ after which every occurrence of $f(s(x), o)$ is replaced by z . Then $\forall \vec{x}. (C \rightarrow D)$ becomes $\forall x, y, u, z, \vec{x} (s(x, y) \wedge o(u) \wedge f(y, u, z) \wedge C' \rightarrow D')$ where C', D' are the result of the substitution in C and D .
- ▶ Example: $a = b$ becomes $a(x) \wedge b(y) \rightarrow x = y$
- ▶ Unicity, e.g., $c(x) \wedge c(y) \rightarrow x = y$, **not** required! (since the new conditions only occur in negative positions)

Puzzle, formalized in CL with functions (Nivelle)

- ▶ Can one color each $n \in \mathbb{N}$ either red or blue but not both such that, if n is red, then $n+2$ is blue, and if n is blue, then $n+1$ is red?
- ▶ No: consider $0?23 \dots$ and $01?34 \dots$
- ▶ CL theory:
 1. $r(x) \vee g(x)$
 2. $r(x) \wedge g(x) \rightarrow \perp$
 3. $r(x) \rightarrow g(s(s(x)))$
 4. $g(x) \rightarrow r(s(x))$
- ▶ Do we miss something?
- ▶ Yes, domain non-empty:
 5. $\exists x. \top$

Puzzle, function eliminated

- ▶ See `LABresources/hdn.in`

1. $r(x) \vee g(x)$
2. $r(x) \wedge g(x) \rightarrow \perp$
3. $r(x) \wedge s(x, y) \wedge s(y, z) \rightarrow g(z)$
4. $g(x) \wedge s(x, y) \rightarrow r(y)$
5. $\exists x. \top$
6. $\exists y. s(x, y)$

- ▶ Solution of version of puzzle with the function:

- ▶ Note that the substitution principle is valid
- ▶ Substitute $(s(x) = y)$ for $s(x, y)$ in 3,4,6:
 - ▶ Regarding 6, $\exists y. s(x) = y$ is trivial
 - ▶ Regarding 4, $g(x) \wedge s(x) = y \rightarrow r(y)$ is equivalent to $g(x) \rightarrow r(s(x))$
 - ▶ Similarly for 3 (and, in general, for any function)

Depth-first proof search in CL

- ▶ Recall the tree construction on slide 8
- ▶ Any open leaf is fine, so we always take the leftmost
- ▶ What instance of which Γ -false axiom to pick?
- ▶ NB two trees: derivation tree and the search space organized as a tree
- ▶ Depth-first search: pick always the first Γ -false axiom from the list, and use the 'simplest' ('oldest') instance
- ▶ Obviously incomplete, but often OK with favourable ordering of coherent axioms:
 1. Facts first, then Horn clauses ($\rightarrow$ goal first)
 2. Disjunctive clauses (cause branching)
 3. Existential axioms (cause new variables)
 4. Disjunctive existential axioms (cause both, the worst)
- ▶ Example: $\exists y. s(x, y)$ should never be put first!

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% the diamond property is preserved under reflexive closure

name(dpe).  :- dynamic e/2,r/2,re/2.

% domain elements a,b,c
dom(a). dom(b). dom(c).

_ axiom assump : (true => re(a,b),re(a,c)).
_ axiom goal_ax(X) : (re(b,X),re(c,X) => goal).

% equality axioms, insofar needed
_ axiom ref_e(X) : (dom(X) => e(X,X)).
_ axiom sym_e(X,Y) : (e(X,Y) => e(Y,X)).
_ axiom congl(X,Y,Z) : (e(X,Y),re(Y,Z) => re(X,Z)).

% intro and elim axioms for re
_ axiom e_in_re(X,Y) : (e(X,Y) => re(X,Y)).
_ axiom r_in_re(X,Y) : (r(X,Y) => re(X,Y)).
_ axiom e_or_r(X,Y) : (re(X,Y) => e(X,Y);r(X,Y)).

_ axiom dp(X,Y,Z) : (r(X,Y),r(X,Z) => dom(U),r(Y,U),r(Z,U)).

```

Breadth-first proof search in CL

- ▶ Recall: Γ is the condition of the leaf node at hand
- ▶ Breadth-first search: collect all 'simplest' instances of Γ -false axioms and use them exhaustively
- ▶ Breadth-first search: complete, but often infeasible
- ▶ With only constants, depth-first complete for forms 1 and 2
- ▶ Depth-first search not complete for **one single existential clause**, subtle:
$$p(a) . \quad p(b) . \quad q(b) \rightarrow \text{goal} .$$
$$p(X) , p(Y) \rightarrow \text{dom}(U) , p(U) , q(X) , r(Y) .$$
- ▶ Wanted: fair application of axioms of form 3 and 4 (sl. 21)
- ▶ Cycling depth-first: depth-first for forms 1 and 2, plus cycling through the (disjunctive) existential clauses, using instances with the 'oldest' constants first. Complete.

Automated reasoning in CL, conclusions

- ▶ Good start: Newman's Lemma (Bezem & Coquand,'03)
- ▶ Limited success in CASC: 50% in FOF (Geo, Nivelle'06)
- ▶ Readable proofs can be extracted from CL proofs
- ▶ Highlight: Hessenberg's Theorem (B, Hendriks, JAR'08)
- ▶ Promising: using SAT techniques (Janicic et al.)
- ▶ A case study, if time allows: Newman's Lemma, stating that, for any strongly terminating relation $r(x, y)$, if r is locally confluent, then r is confluent. Informal proof on blackboard, code in `nl.in`. Many interesting aspects.

Proof assistants

- ▶ In proof assistants, proof objects are required
- ▶ CL proofs are readable and easily convertible
- ▶ Provers outputting proof objects:
 - ▶ `cl.pl` (B, exports proofs to Coq, also used to verify them)
 - ▶ `coherent` (Isabelle tactic, Berghofer)
 - ▶ ArgoCLP (Coq, Isar, natural language)
- ▶ Modern automated support of proof assistants centers around specialized tools for decidable fragments of FOL, using SAT Modulo Theories-techniques. Very useful is, e.g., the tactic `lia` (linear arithmetic) in Coq.

Model finding

- ▶ Satisfiability in FOL is co-RE, so restrict to finite models
- ▶ Naive approach: try to find a model with 1 element, then with 2 elements, and so on. Quantifiers $\forall, \exists$ are written out ('grounding'), and the resulting (rapidly growing) propositions are fed to a SAT-solver
- ▶ Many clever tricks can actually make this to work
- ▶ CL proof search is not finite model complete: $\exists y. s(x, y)$
- ▶ Solution (Nivelle): use (exhaustively) old constants before you generate a new one + use lemma learning
- ▶ Success in CASC'07: 81% in FNT (Geo, Nivelle) (Paradox, based on Minisat, winner with 85%)
- ▶ CL competitive on problems 'too big to ground'

Constructive algebra

- ▶ Pioneers of applying CL/GL to constructive algebra:
Coste, Lombardi, Roy, Coquand
- ▶ Idea: making constructive sense of classical proofs by exploiting that significant parts of algebra can be formalized in CL/GL
- ▶ Barr's Theorem guarantees then that classical results are provable in CL/GL

Algebraic theories in CL/GL

- ▶ Ring (commutative with $1 \neq 0$): equational axioms
- ▶ Local ring: $\exists y. (x \cdot y = 1) \vee \exists y. ((1 - x) \cdot y = 1)$
- ▶ Field: $(x = 0) \vee \exists y. (x \cdot y = 1)$ (makes = decidable!)
- ▶ Alg. closed: $\exists x. x^{n+1} = a_0 + a_1x + \cdots + a_nx^n$ (all $n \in \mathbb{N}$), so infinitely many coherent axioms
- ▶ Positive formula using an infinite disjunction:
 $\bigvee_{n \in \mathbb{N}} 0 = x^{n+1}$, expressing that x is nilpotent

Hilbert's Nullstellensatz

- ▶ Consider fields $k \subset K$ with K algebraically closed. Let I be an ideal of $k[\vec{x}]$, and $V(I)$ the set of common zeros (Nullstellen) in K of the polynomials in I . Then: for any $p \in k[\vec{x}]$ such that p is zero on $V(I)$ there exists an n such that $p^n \in I$.
- ▶ Example: $\mathbb{Q} \subset \mathbb{C}$, $I = (x^4 + 2x^2 + 1)$, $p = x^5 - x$, $p^2 \in I$
- ▶ In its full generality, Hilbert's Nullstellensatz is a strong classical theorem, with lots of special cases and variations
- ▶ Effective Nullstellensatz: **compute** the n such that $p^n \in I$
- ▶ Dynamical method in algebra: Effective Nullstellensätze, Coste, Lombardi, Roy, 2001 (Dynamic method = CL proof)